

20. Write a note on ROM – its uses and types.

21. Explain the role of a cache memory chip in a computer.

Read Only Memory (ROM)

- § ROM a non-volatile memory chip
- § Data stored in a ROM can only be read and used – they cannot be changed
- § ROMs are mainly used to store programs and data, which do not change and are frequently used. For example, system boot program

Types of ROMs

Type	Usage
Manufacturer-programmed ROM	Data is burnt by the manufacturer of the electronic equipment in which it is used.
User-programmed ROM or Programmable ROM (PROM)	The user can load and store "read-only" programs and data in it
Erasable PROM (EPROM)	The user can erase information stored in it and the chip can be reprogrammed to store new information

Types of ROMs

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Type	Usage
Ultra Violet EPROM (UVEPROM)	A type of EPROM chip in which the stored information is erased by exposing the chip for some time to ultra-violet light
Electrically EPROM (EEPROM) or Flash memory	A type of EPROM chip in which the stored information is erased by using high voltage electric pulses

Cache Memory

- § It is commonly used for minimizing the memory-processor speed mismatch.
- § It is an extremely fast, small memory between CPU and main memory whose access time is closer to the processing speed of the CPU.
- § It is used to temporarily store very active data and instructions during processing.

Cache is pronounced as "cash"

22. Define following terms :

- a. Hardware
- b. Software
- c. Firmware
- d. Middleware

23. Explain the types of software giving examples of each.

Hardware isn't given in notes! I'll add as soon as I get!



- § *Hardware* refers to the physical devices of a computer system.
- § *Software* refers to a collection of programs
- § *Program* is a sequence of instructions written in a language that can be understood by a computer
- § *Software package* is a group of programs that solve a specific problem or perform a specific type of job

Firmware

- § Firmware is software substituted for hardware and stored in read-only memory
- § Firmware technology has enabled production of various types of smart machines having microprocessor chips with embedded software

Middleware

- § Basic idea is to have a *separate software layer* to:
 - § Act as “glue” between client and server parts of application
 - § Provide programming abstraction
 - § Mask heterogeneity of underlying network, hardware, and OS
- § Encourages *three-tier* software architecture against two-tier popularized by Server-Client architecture

Types of Software

Most software can be divided into two major categories:

- § *System software* are designed to control the operation and extend the processing capability of a computer system
- § *Application software* are designed to solve a specific problem or to do a specific task

System Software

- § Make the operation of a computer system more effective and efficient
- § Help hardware components work together and provide support for the development and execution of application software
- § Programs included in a system software package are called *system programs* and programmers who prepare them are called *system programmers*
- § Examples of system software are operating systems, programming language translators, utility programs, and communications software

Application Software

- § Solve a specific problem or do a specific task
- § Programs included in an application software package are called *application programs* and the programmers who prepare them are called *application programmers*
- § Examples of application software are word processing, inventory management, preparation of tax returns, banking, etc.

24. Explain following terms :

- a. Compiler
- b. Assembler
- c. Interpreter
- d. Linker
- e. Source program
- f. Object program
- g. Macro instruction

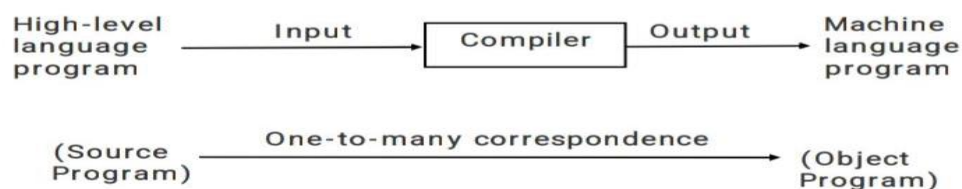
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25. Discuss types of computer languages with advantages and limitations of each

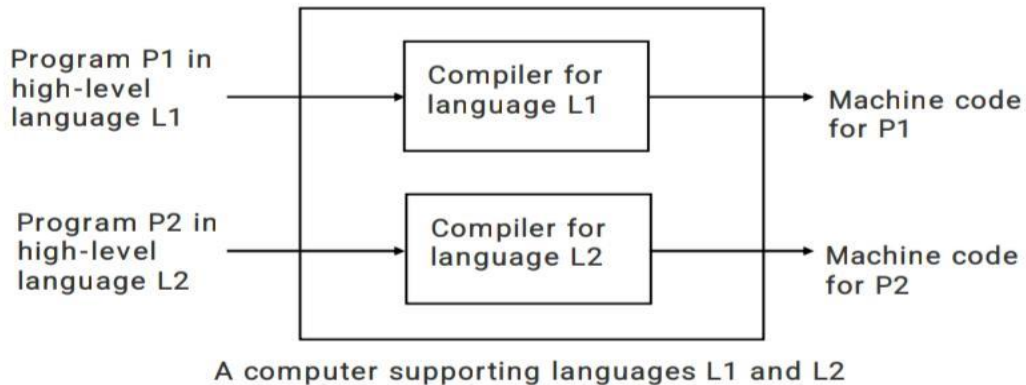
Compiler

- § Translator program (software) that translates a high-level language program into its equivalent machine language program
- § Compiles a set of machine language instructions for every program instruction in a high-level language

Compiler



Compiler

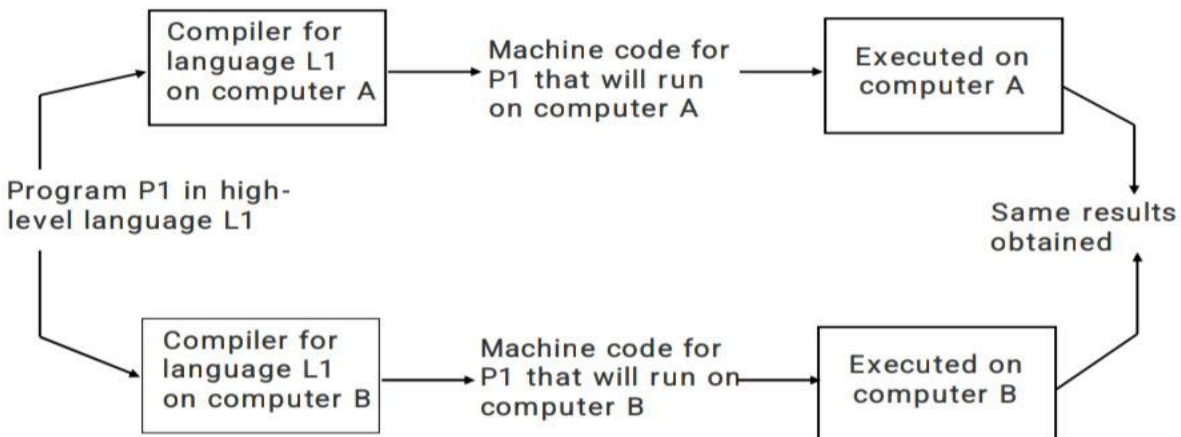


Illustrating the requirement of a separate compiler for each high-level language supported by a computer

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Compiler

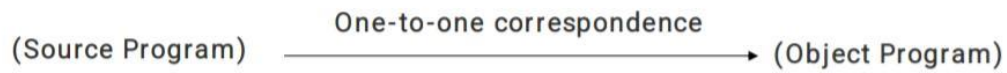
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Illustrating the machine independence characteristic of a high-level language. Separate compilers are required for the same language on different computers

Assembler

- § Software that translates an assembly language program into an equivalent machine language program of a computer

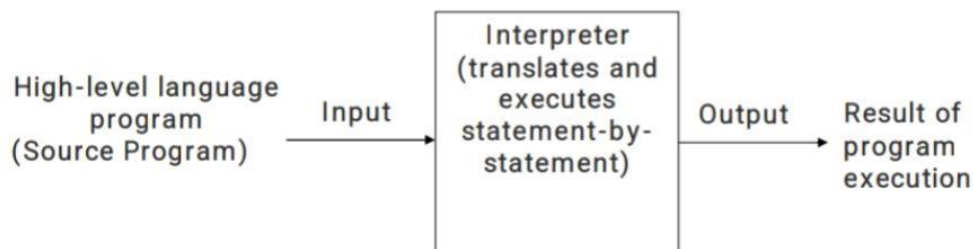


Interpreter

- § Interpreter is a high-level language translator
- § Takes one statement of a high-level language program, translates it into machine language instructions
- § Immediately executes the resulting machine language instructions
- § Compiler simply translates the entire source program into an object program and is not involved in its execution



Role of an Interpreter



Linker

- § For a large software, storing all the lines of program code in a single source file will be:
 - Difficult to work with
 - Difficult to deploy multiple programmers to concurrently work towards its development
 - Any change in the source program would require the entire source program to be recompiled
- § Hence, a modular approach is generally adapted to develop large software where the software consists of multiple source program files
- § No need to write programs for some modules as it might be available in library offering the same functionality

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Linker

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- § Each source program file can be independently modified and compiled to create a corresponding object program file
- § Linker program (software) is used to properly combine all the object program files (modules)
- § Creates the final executable program (load module)

Broad Classification of Computer Languages

- § Machine language
- § Assembly language
- § High-level language

Machine Language

- § Only language of a computer understood by it without using a translation program
- § Normally written as strings of binary 1s and 0s
- § Written using decimal digits if the circuitry of the computer being used permits this

Advantage

- § Can be executed very fast

Limitations

- § Machine Dependent
- § Difficult to program
- § Error prone
- § Difficult to modify

Assembly/Symbolic Language

Programming language that overcomes the limitations of machine language programming by:

- § Using alphanumeric mnemonic codes instead of numeric codes for the instructions in the instruction set
e.g. using ADD instead of 1110 (binary) or 14 (decimal) for instruction to add
- § Allowing storage locations to be represented in form of alphanumeric addresses instead of numeric addresses
e.g. representing memory locations 1000, 1001, and 1002 as FRST, SCND, and ANSR respectively
- § Providing pseudo-instructions that are used for instructing the system how we want the program to be assembled inside the computer's memory
e.g. START PROGRAM AT 0000; SET ASIDE AN ADDRESS FOR FRST

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Advantages of Assembly Language Over Machine Language

- § Easier to understand and use
- § Easier to locate and correct errors
- § Easier to modify
- § No worry about addresses
- § Easily relocatable
- § Efficiency of machine language

Computer Fundamentals: Pradeep K. Sinha & Prit

Limitations of Assembly Language

- § Machine dependent
- § Knowledge of hardware required
- § Machine level coding

High-Level Languages

- § Machine independent
- § Do not require programmers to know anything about the internal structure of computer on which high-level language programs will be executed
- § Deal with high-level coding, enabling the programmers to write instructions using English words and familiar mathematical symbols and expressions

Advantages of High-Level Languages

- § Machine independent
- § Easier to learn and use
- § Fewer errors during program development
- § Lower program preparation cost
- § Better documentation
- § Easier to maintain

Limitations of High-Level Languages

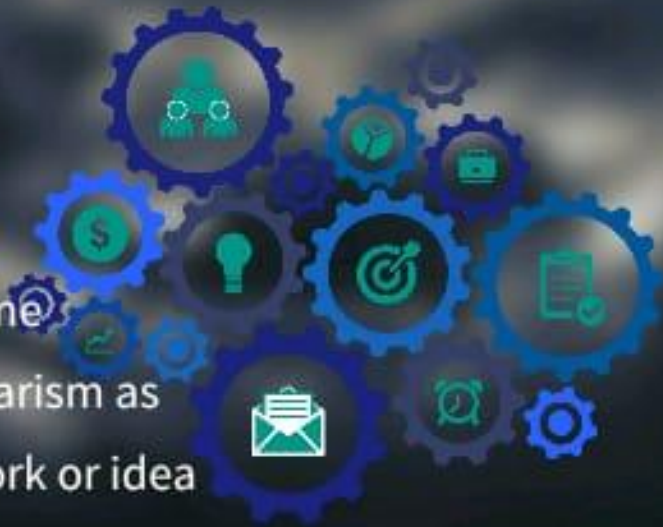
- § Lower execution efficiency
- § Less flexibility to control the computer's CPU, memory and registers

Assignment Questions

1. Explain the term - Plagiarism. Discuss types of Plagiarism.
2. Explain the process of plagiarism checking.

Plagiarism: No Authentic Definition

- ◆ The word plagiarism originates from the Latin word 'plagiarius' (mean
- ◆ 'kidnapper') in the 17th century. Merriam-Webster's Online Dictionary defines plagiarism as "the act of stealing of work or idea or language with out proper



What is Plagiarism?

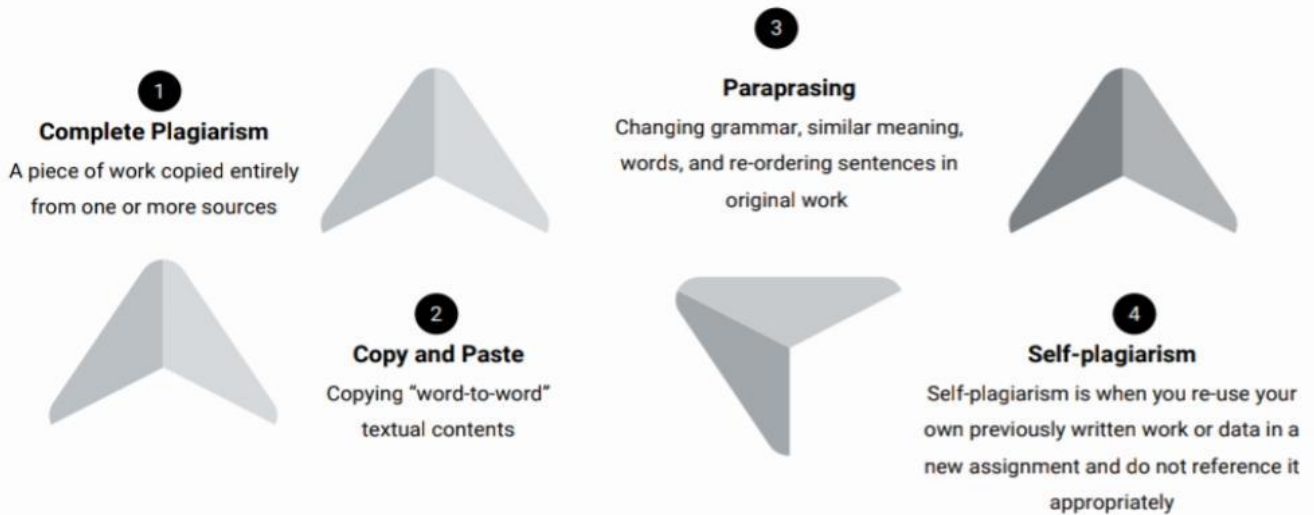
- ◆ to steal and pass off (the ideas or words of another) as one's own;
- ◆ to use (another's production) without crediting the source;
- ◆ to commit literary theft; and
- ◆ to present as new and original an idea or product derived from an existing source.

Plagiarism includes:

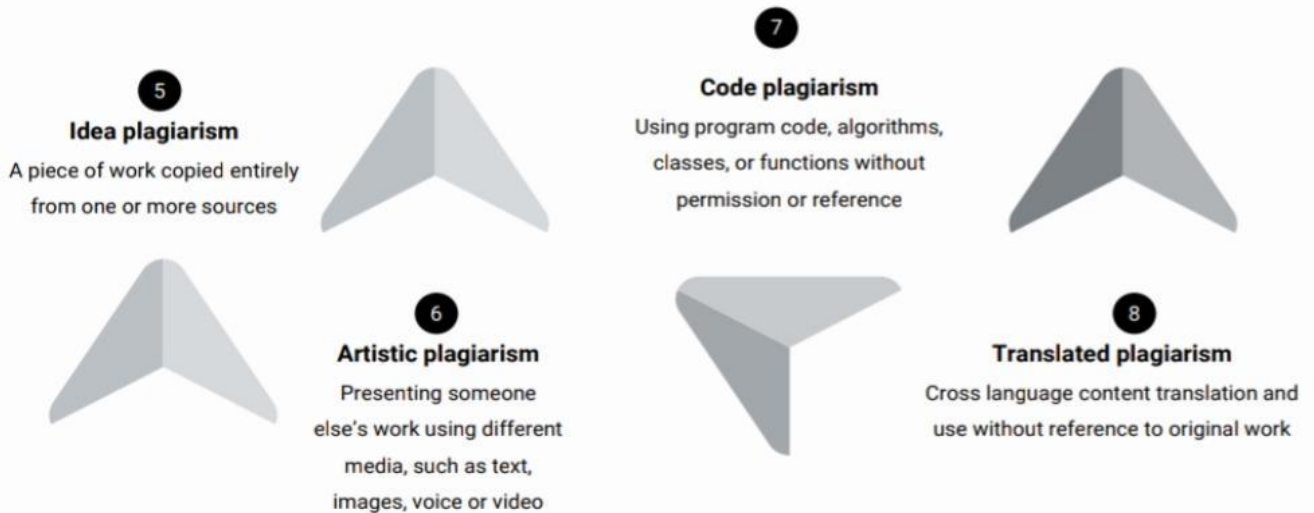
- ① using someone else's exact words without using quotation marks or a citation;
- ② using someone else's ideas without using a citation (even if the source is in the bibliography);
- ③ using paraphrased words from a source without a citation.
- ④ Copying several words or ideas from a source that it makes up the majority of

Using rearranged words from a source without citation

Types of Plagiarism: Intentional / Unintentional



Types of Plagiarism: Intentional / Unintentional



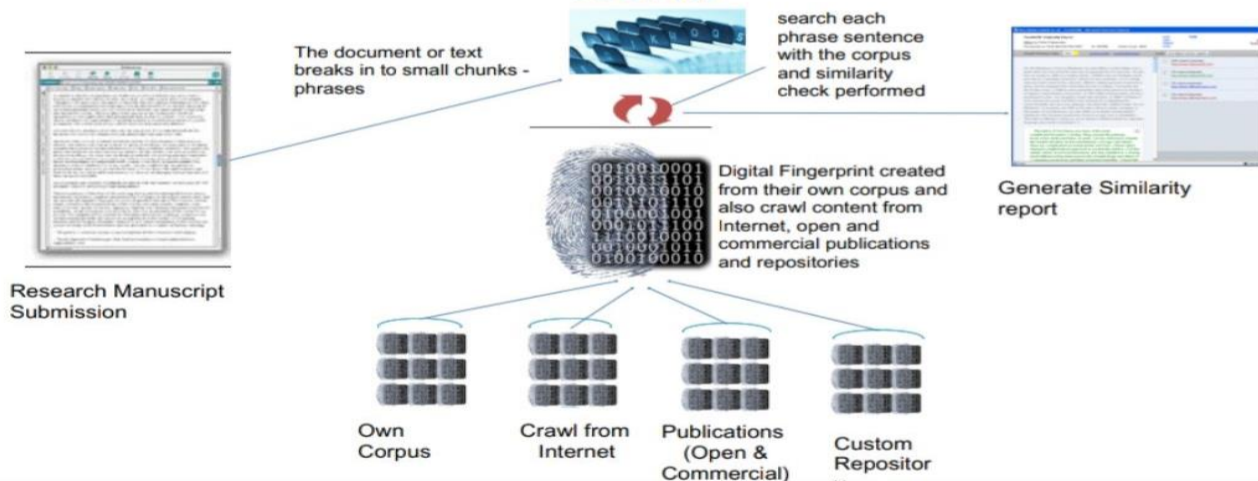
Plagiarism Detection

Computer-assisted Detection: Using Plagiarism detection software

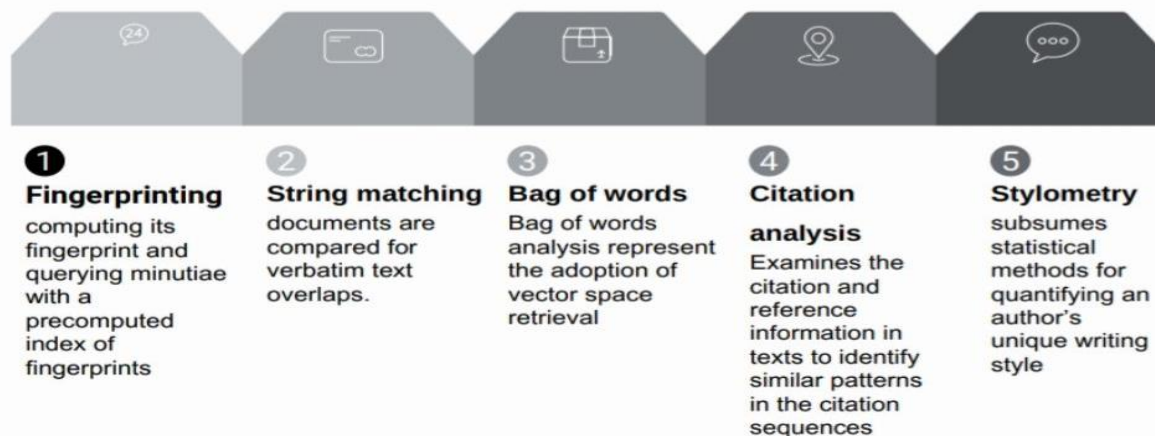
- a set of powerful algorithms is used to create a digital fingerprint of a text document.
- document's fingerprint is cross-referenced against local databases, and Internet.
- a colour-coded originality report, complete with source links is generated.



How does plagiarism checker work?



Similarity Detection Approaches



3. Name the following : Open source anti-plagiarism tools, commercial anti-plagiarism tools.
4. Discuss in brief following anti-plagiarism tools : Viper, DupliChecker
5. Explain limitations of plagiarism checker software

Anti-plagiarism Tools



Open Source

- Viper

www.scanmyessay.com

- Dupli Checker

www.dupliChecker.com

- Plagiarism Checker

<http://www.plagiarismchecker.com>

- Quetext

<https://www.quetext.com/>

Commercial Products

- Turnitin

www.turnitin.com

- iThenticate

www.ithenticate.com

- URKUND

www.urkund.com/

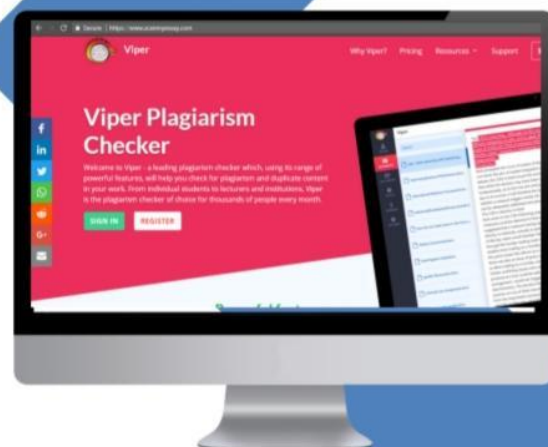
Viper

Pros

- 100% Free
- All features are available to everyone on the Internet.
- Your document will be checked against more than 10 billion online resources including journals, books and websites.

Cons

- Mostly targeted to academic students as it is widely used for evaluating student papers.



Dupli Checker

Pros

- Free of charge
- Ease of use
- 2 Ways to check for Plagiarism
Either copy and paste your text in the field and then check it for plagiarisms, or upload a Docx or Text file from your computer.
- 50 Plagiarism scans per day for registered users

Cons

- 1 Search per day for unregistered users



Limitations of Plagiarism Detection Software

- Commercially available software very costly
- Limitation of database available for cross checking
- Time consuming to check plagiarism using multiple sources with human intervention.
- Original research work and data are made available to a third party which may lead to copyright violation.
- Plagiarism detection software can be fooled by making intelligent changes in syntax of plagiarized portion of article or assignment.
- Detects common phrases and sentences that are used often used by everyone leading to increase in overall proportion of plagiarized content in a given text.

6. Explain role of utility software with examples

7. Describe use of defragmentation software and compression utility

Utility Software



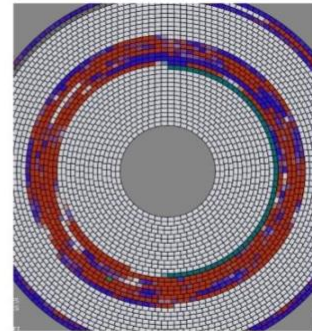
- Utility programs are designed to do just one or two tasks
- Utility software interacts with the computers hardware
 - E.g. the Disk Drive,
- Utility programs are in built in to the Operating System or could be bought/downloaded as separate programs
- Utilities are not essential for the computer to work, but either make it easier for the user to use or do something, or provide some “housekeeping”

Utility Programs

- Disk/File Management eg Disk Copying, Disk Formatting
- Disk Defragmentation
- File Compression
- File Repair
- Disk Backup / Restore
- Security utilities eg Anti-Virus, Firewall
- Encryption

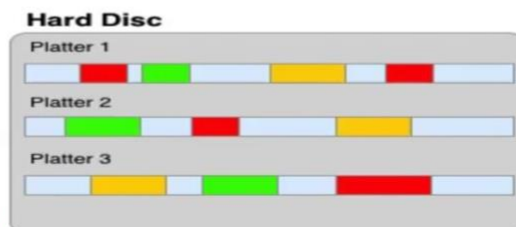
Defragmentation Software

- When a hard disk drive is new – files get added onto the disk in order – very much like starting with a blank piece of paper and adding to it
- As files are deleted – this leaves gaps
- When new files are saved – the files fill the gaps and become fragmented
- Defragmentation software groups fragmented files back together



Defragmentation Software

- Here we have a hard disc with 3 platters.
- We have 3 files but they are not together so the read/write head needs to move to different places to read the whole file. This is slow.
- Defragmenting a drive moves the file fragments so that they are all in continuous sectors.
- This means the read/write head does not have to move as far and so can read files faster.
- The defragmenter also groups all the free disk space together so that new files can be store in one place
- This optimises disk performance



Compression

- Reducing the size of the file by performing an algorithm on the original data to remove redundant data
- **Lossless**
 - The original file can be re-created as no data is lost.
 - For example: used on text files where it has to be possible to restore the files exactly otherwise you wouldn't be able to read them
- **Lossy**
 - Some of the original data is lost and the original file can not be re-created
 - For example: a photo where some detail may be lost but this will be undetectable to the human eye on screen

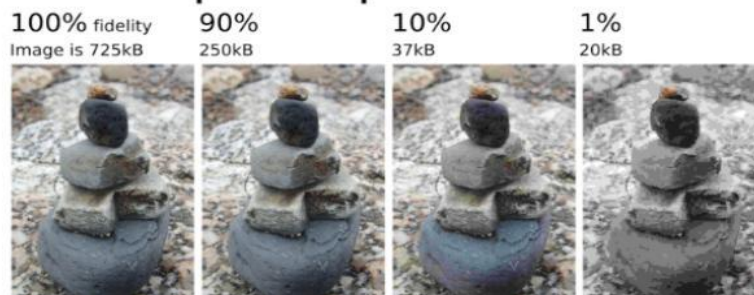
GCSE (9–1)
COMPUTER SCIENCE

By the end of this lesson, you will understand:

- Utility system software:
- The role and methods of backups

Why compress files?

- ✓ Zipped or compressed files can be transmitted much more quickly over the internet
- ✓ Sometimes there is a file size limit – compressing can help keep your file within the limit
- ✓ Download and upload speeds are reduced



8. Discuss following utilities : Disk checking, Disk clean-up, Encryption software, Firewall, Antivirus software
9. Write a note on Backup and archiving techniques in computer systems

Other utilities

- The operating system will come with disk organisation and management tools - for example move files, copy files, rename files.
- You can get utility programs to carry out other functions on the disk:
 - **Disk checking and repairing:** locate and solve problems such as bad links between file fragments and identifies bad sectors on the disk, keeping a record of where the problems are located. Bad sectors can then be avoided in future
 - **Disk/System clean-up** – identifies temporary files, cached copies of files and unused files that can be removed safely leaving more free space on the hard disk

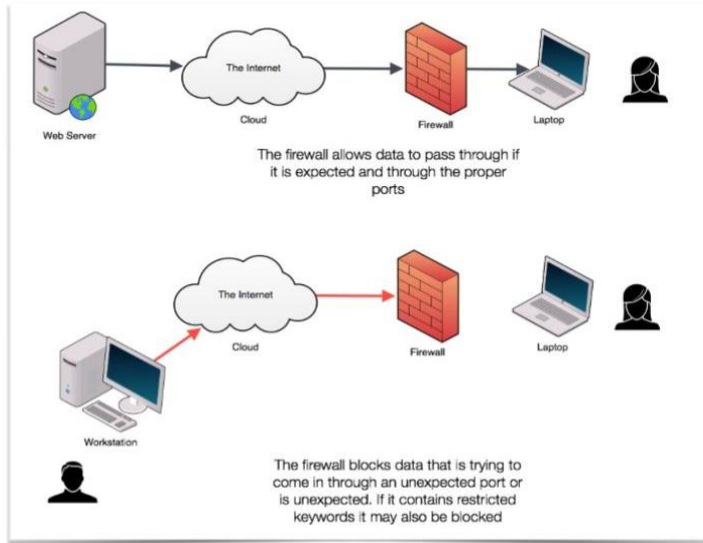
Encryption Software

ACCESS:

- Keeping data secure is important for data protection reasons as well as being worth money for a business.
- Sensitive data can be encrypted so that a password (or other security like a fingerprint scanner) must be used to be able to read the files.
- Encryption means to scramble data in a way that it is unreadable to anybody who doesn't have a key to be able to unscramble this.
- Data is encrypted using a key and then decrypted using a key
- The encryption process uses an algorithm to transform plain text into cipher text
- To decode the original information, you would need to know the algorithm and the key

Security Utilities

Firewall



- A firewall is software that monitors data coming into the computer from a network.
- It prevents data that is not arriving in the correct manner from coming in.

Security Utilities

Anti-virus & Anti -Malware Software



- Viruses, Trojan horses, and worms are malicious software programs.
- They can be written to disrupt your computer, and even enable hackers to take it over.
- Malware includes programs such as spyware, adware, key loggers.
- Anti virus/malware software works by
 - Scanning files on your computer for suspicious activity.
 - Detects, prevents and removes malware by scanning incoming files
 - Safety removes anything that is considered a threat

Backups



- Backups are completed to ensure there is a way of restoring files/data/programs in case of computer failure
- Copies of data and/or software are taken regularly and usually stored in a different location
 - Ensures that if computers are stolen or damaged in for example a fire or a natural disaster that data can be restored and the organisation doesn't lose its data completely. Depending on the organisation, the implications of losing data could be very expensive!

Backups



- In some organisations, like a school, a daily backup may be sufficient
- But in others such as a bank or retail company, every transaction needs to be duplicated to a second or even third storage device in a remote location
- Various options:
 - ☐ **How much?** back up important files or back up whole drive,
 - ☐ **Which device?** Use an external hard drive or other device that is portable and high capacity stored off site or fire proof safe.
 - ☐ **Or,** use remote data storage such as the cloud where your data is safely saved in a huge data warehouse, with data stored at multiple sites.

Archiving

- Often there is a large amount of data stored on computer that is no longer needed on a regular basis eg student data from 2009
- You can't delete it in case it is needed again, and in business, companies are required to keep data for a number of years for tax purposes
- This can be taken off the main system and stored in an archive, often on magnetic tape as it is cheap.
- This frees up space for data used more regularly/recently.

